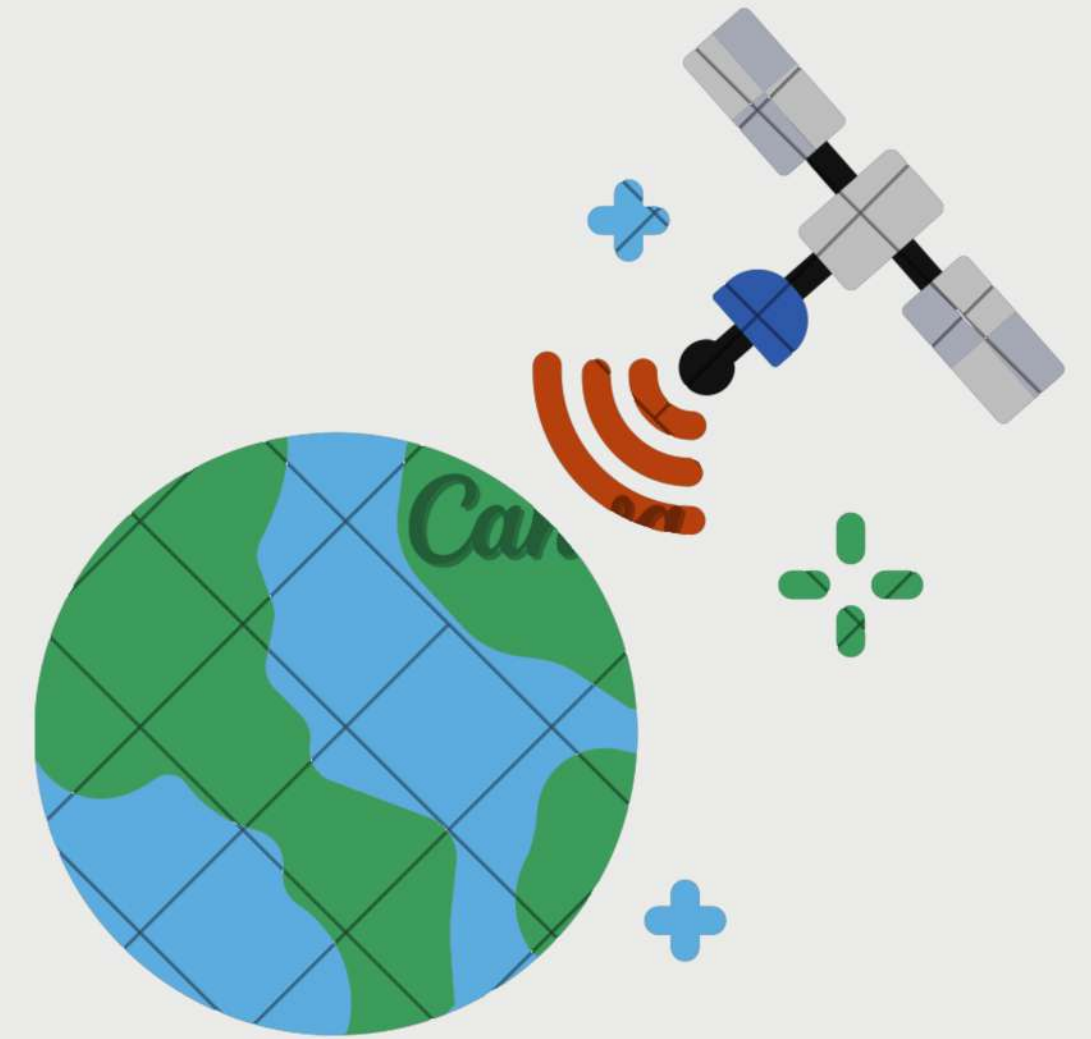
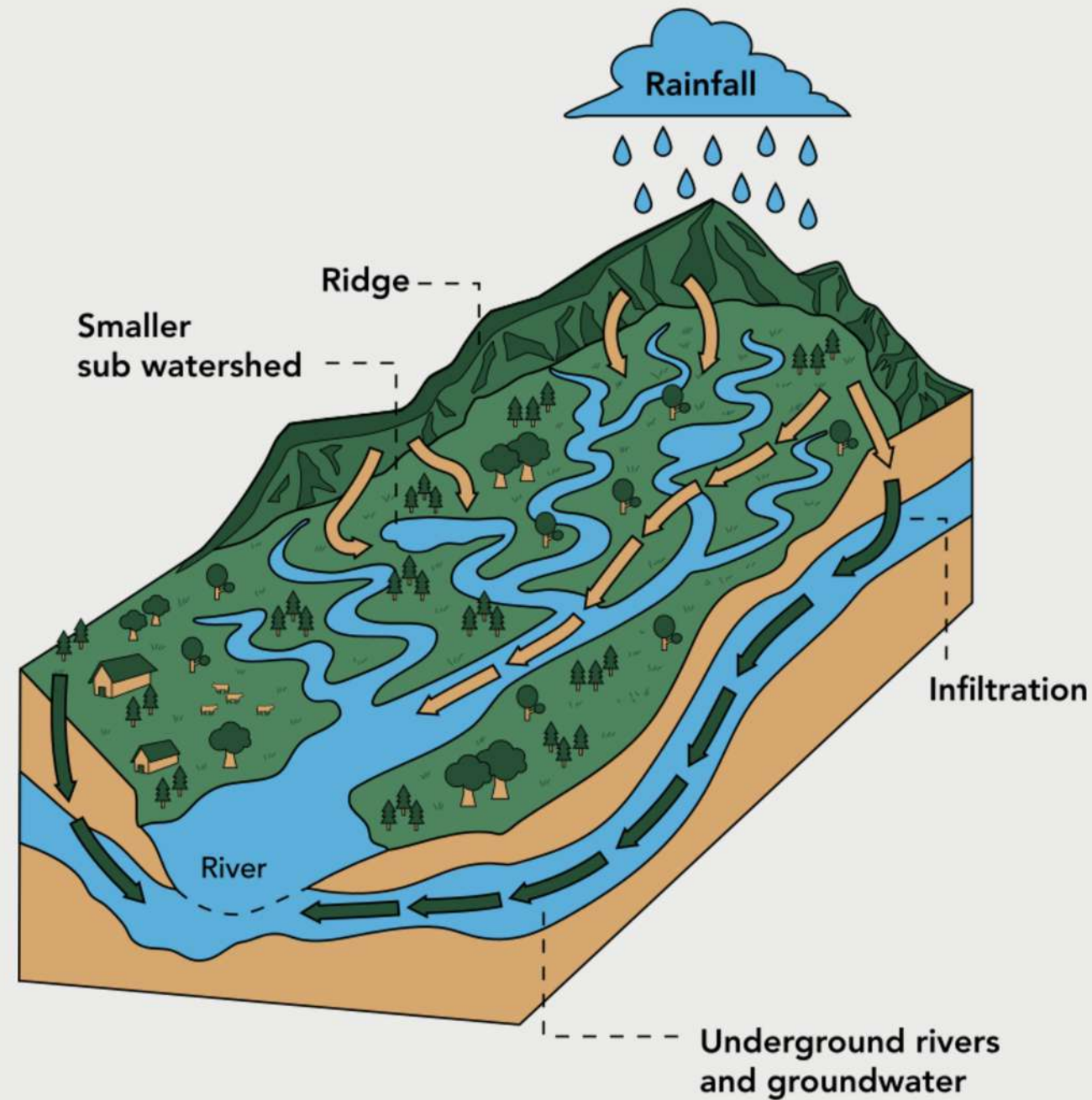


Morphometric analysis and prioritization of sub-watersheds in the ghataprabha basin using GIS and Remote Sensing techniques.



Introduction :-

Effective water resource management is more critical than ever. In complex river systems like the Ghataprabha Basin, traditional manual field surveys often fall short due to time and scale limitations. That is where modern geoinformatics comes in.

By leveraging Geographic Information Systems (GIS) and Remote Sensing (RS), we can unlock a deeper, data-driven understanding of our vital hydrological ecosystems.

I am Using :-

1) Basic Morphometric Parameters

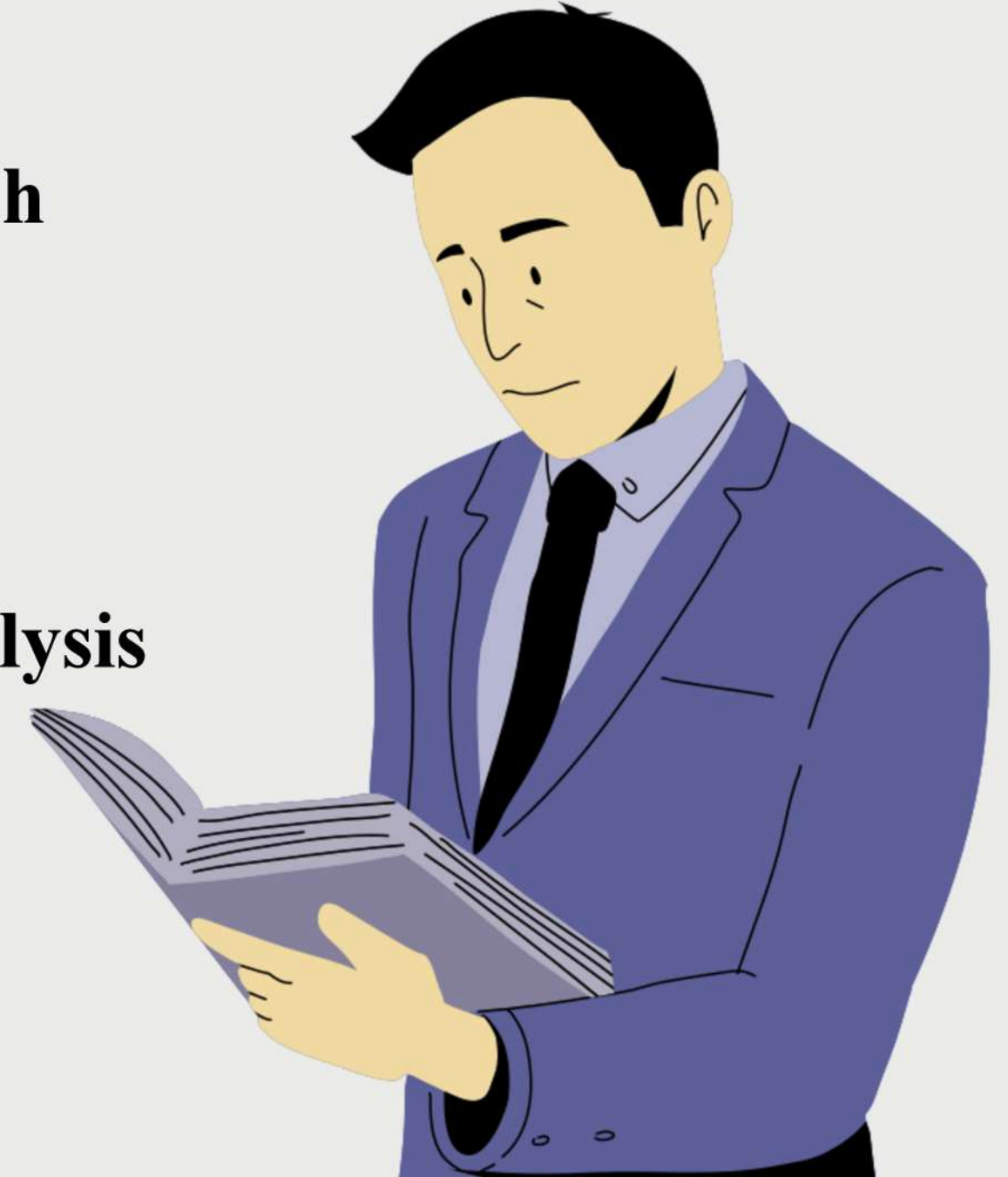
2) Linear Aspects

3) Areal Aspects

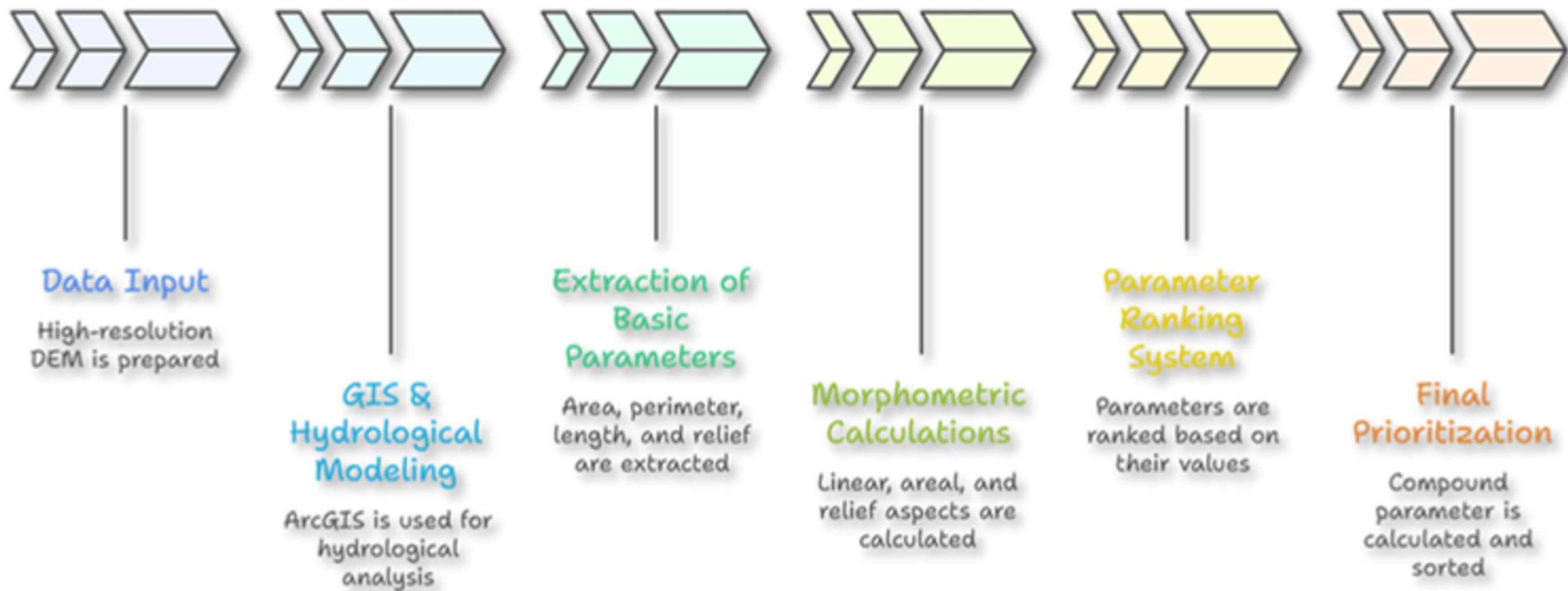
4) Relief Aspects

Objective

- 1) To delineate the study area's watershed and sub-watersheds using geospatial techniques.**
- 2) To analyze and calculate the morphometric parameters of each sub-watershed using geospatial techniques.**
- 3) To prioritize the sub-watersheds based on morphometric analysis**



Morphometric Analysis and Prioritization Methodology



Data use :-

Digital Elevation Model (DEM):

SRTM (Shuttle Radar Topography Mission) 30-meter resolution DEM was downloaded from USGS Earth Explorer

GIS Software:

ArcGIS Pro (version 2.9) was used for spatial analysis, watershed delineation, and map generation. Arc Hydro Tools extension was utilized for hydrological analysis and stream network extraction.

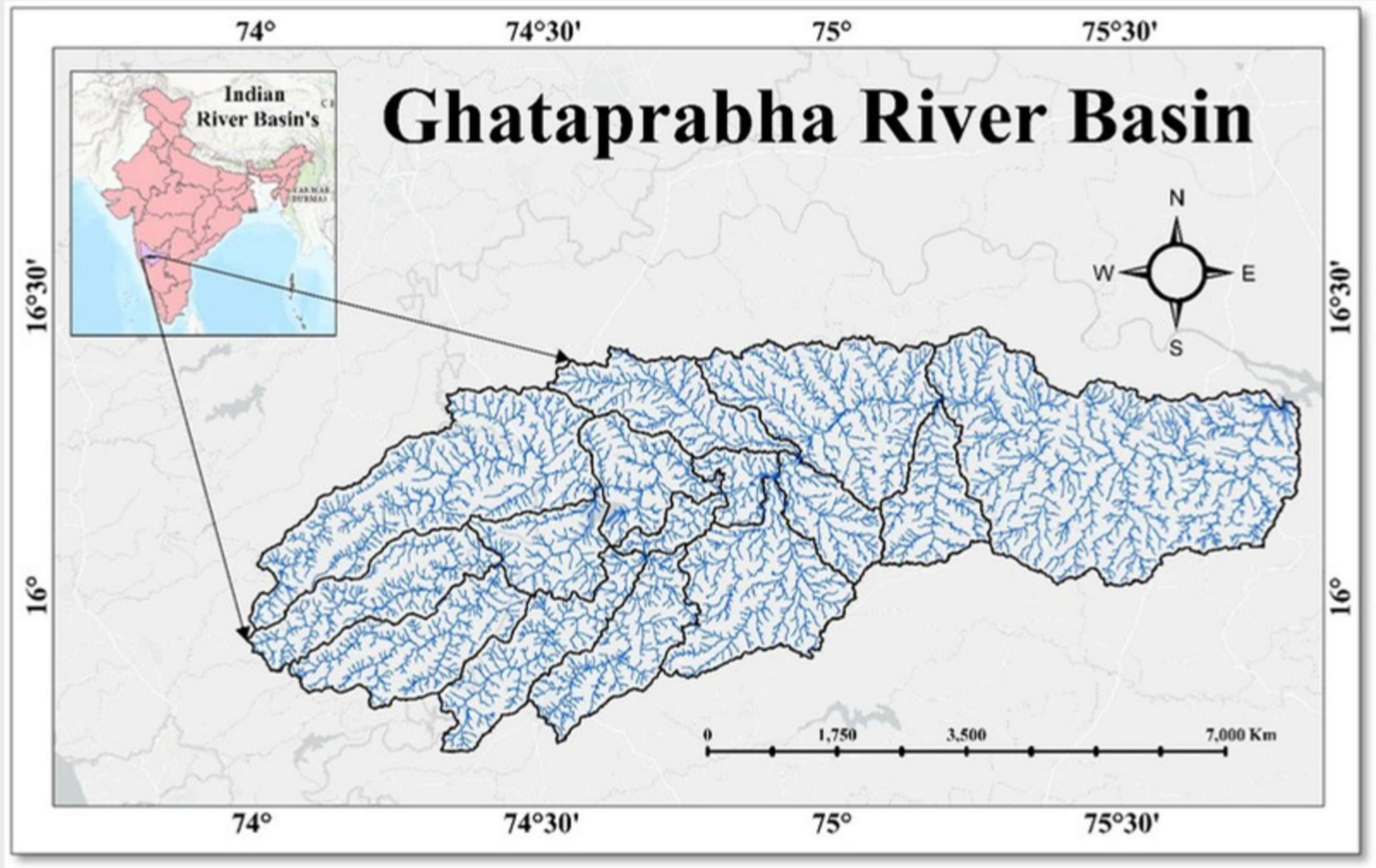
SWAT (Soil and Water Assessment Tool):

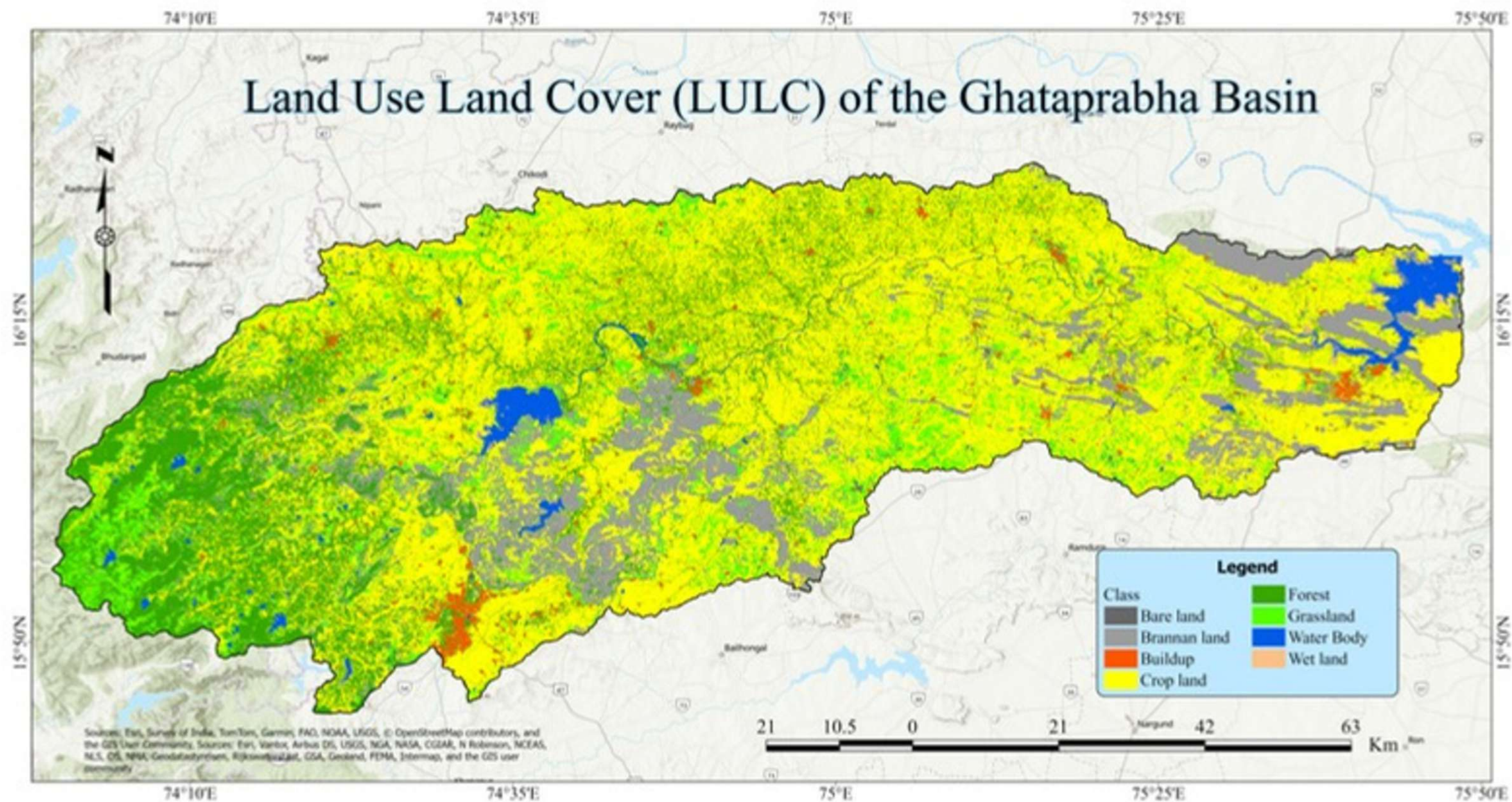
ArcSWAT was employed for automated watershed delineation and sub-watershed boundary definition.

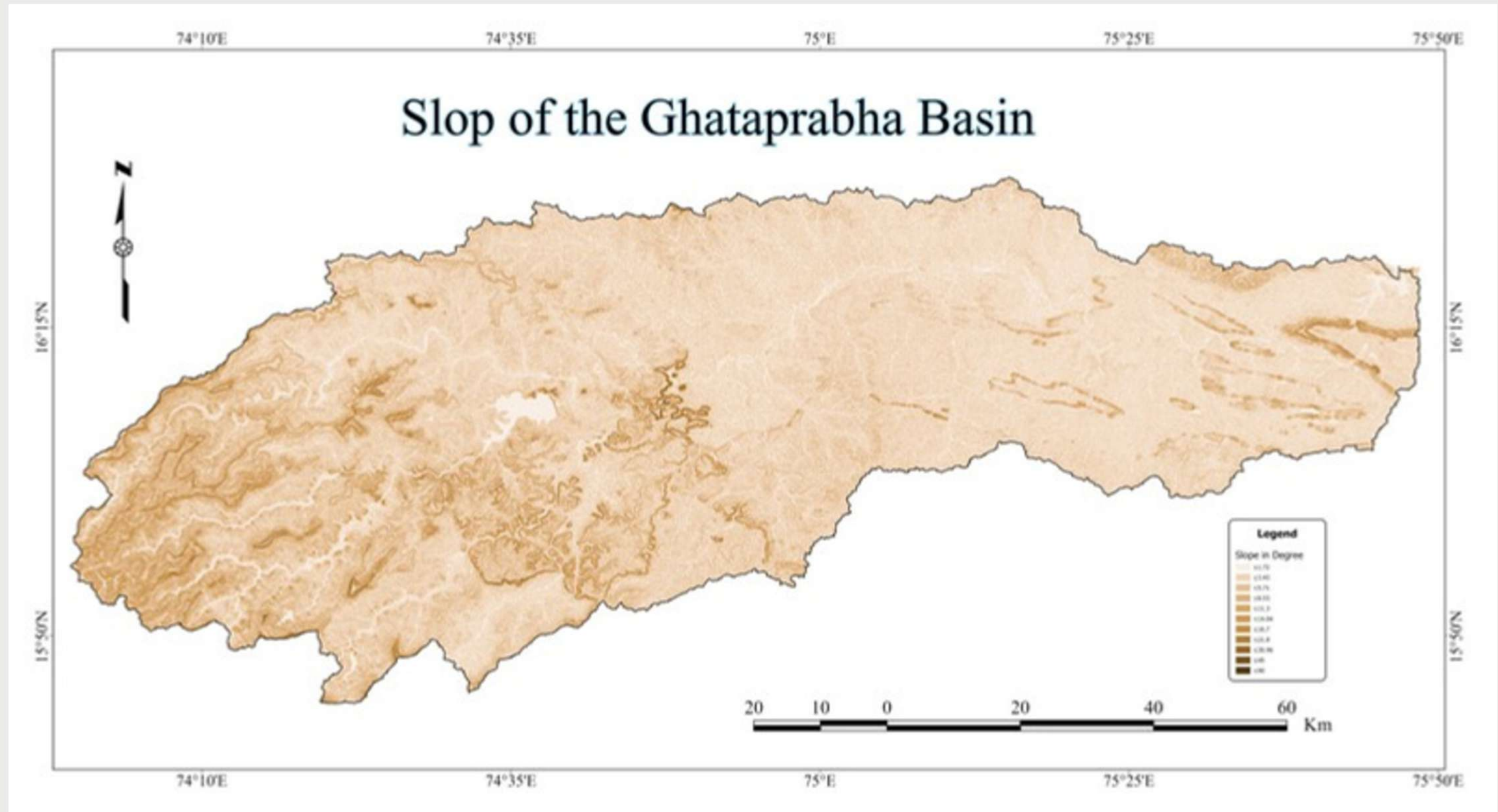
Satellite Imagery:

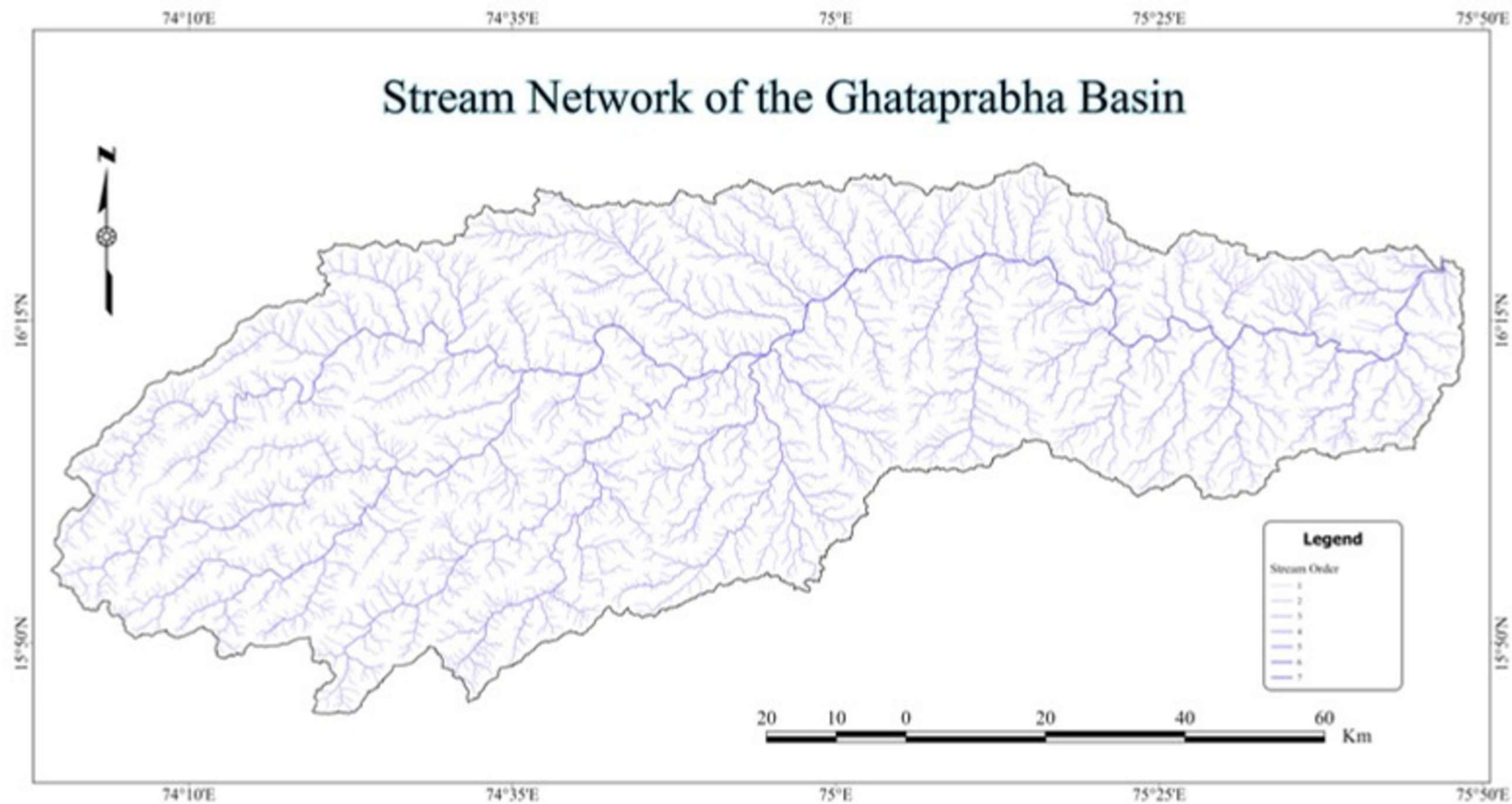
High-resolution Landsat-8 imagery (30-meter resolution) was used for Land Use/Land Cover (LULC) classification and visual validation of the analysis.

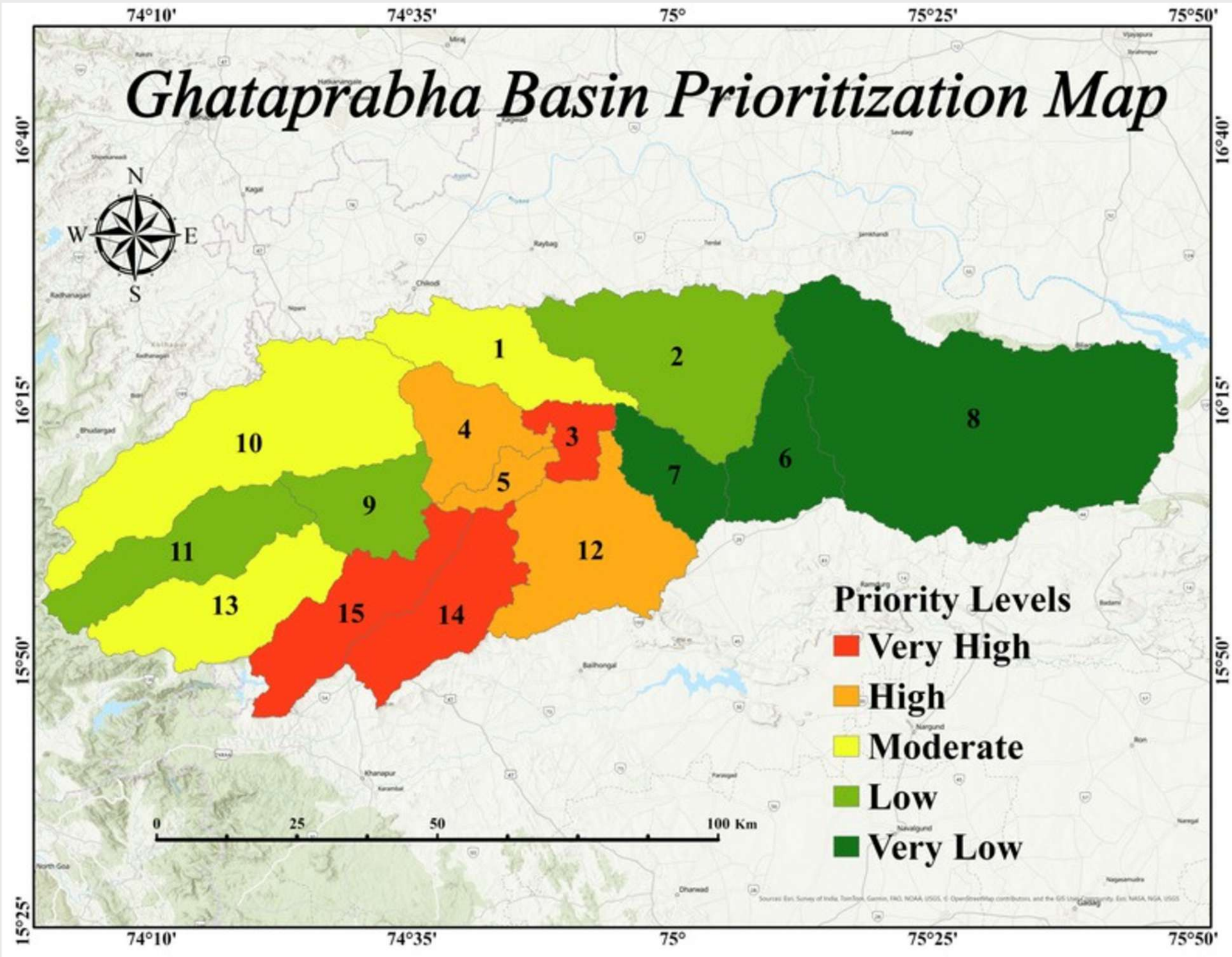
Maps:-











Conclusion :-

This study has demonstrated the utility of GIS and Remote Sensing in conducting a comprehensive morphometric analysis of the Ghataprabha Basin and in generating a scientifically robust sub-watershed prioritization framework. The fifteen delineated sub watersheds exhibit considerable variation across all three categories of morphometric parameters—linear, areal, and relief—reflecting the diverse physiographic and geological character of the basin. SW14 and SW15 stand out as the most critical sub-watersheds requiring immediate conservation attention on account of their extreme relief and ruggedness values, while SW3 and SW5 demand urgent intervention due to their high drainage density, stream frequency, and bifurcation ratios. The Prioritization Map produced through this analysis provides an objective, evidence based tool for planners and conservation agencies to allocate limited resources strategically and to implement soil and water conservation measures where they are needed most. The methodology adopted is transferable to other river basins and can be updated as new satellite data becomes available, ensuring its continued relevance for long-term watershed management and environmental stewardship in the region.